

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of M. Pharm (Pharmaceutical Technology)

University Theory Examination Dec 2016

PHPT002, Pharmaceutical Product Development

Date: 19.12.2016 , Monday

Time: 10:00 a.m to 1:00 p.m

Maximum Marks: 70

Instructions:

1. Q.1 and Q.4 are compulsory.
2. Figures to right indicate marks.
3. Section wise separate answer book to be used.
4. Draw neat sketches wherever necessary.

SECTION-I

- Q.1** **A)** What is shelf life? How shelf life is determined? **05**
 B) Write a note on: Accelerated Stability Study **05**
- Q.2** **A)** What is Dissolution and Solubility? Enlist theories of dissolution and describe about diffusion layer model. **07**
 B) Enumerate various factors affecting rate of dissolution. **03**

OR

- Q.2** **A)** Discuss the factors affecting selection of dissolution medium. **05**
 B) Write a note on: similarity factor and dissimilarity factor **05**
- Q.3** **Answer the following (Any three)** **15**
 (A) Enlist physical, chemical & pharmaceutical factors affecting preformulation. Discuss chemical decomposition factors with special reference to photodecomposition and methods to retard it.
 (B) Describe about bracketing and Matrixing designs.
 (C) What is ICH guidelines and what are its objectives? Enlist the specific guidelines under Q1.
 (D) Enlist different chemical degradation pathways in preformulation studies? Explain the various factors influencing degradation pathway?

SECTION-II

- Q.4** Write a short note on superdisintegrant and co processed excipients. **10**
- Q.5** **(A)** Give Classification of biodegradable polymers with examples. **05**
 (B) Give Application of polymers. **05**

OR

- Q.5** Application of phase rule to the characterization of polymorphic systems. **10**

Q.6 **Answer the following (Any three)**

15

- (A) Explain: Types of polymorphs.
- (B) Discuss the impact of polymorphism on quality of lyophilized products.
- (C) Factors affecting selection of excipients.
- (D) Mechanism of biodegradation of polymer in the body.